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// 30-MINUTES PLENARY TALK

Reinforcement Learning in Continuous Representation Spaces

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Robotics

Areas

- Mobile Robotics
- Bio-inspired Robotics
- Cognitive Robotics
- Intelligent Robotics
- Human-Robot Interaction
- Tele-operated Robotics
- Swarm Robotics

...



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Perception and Action

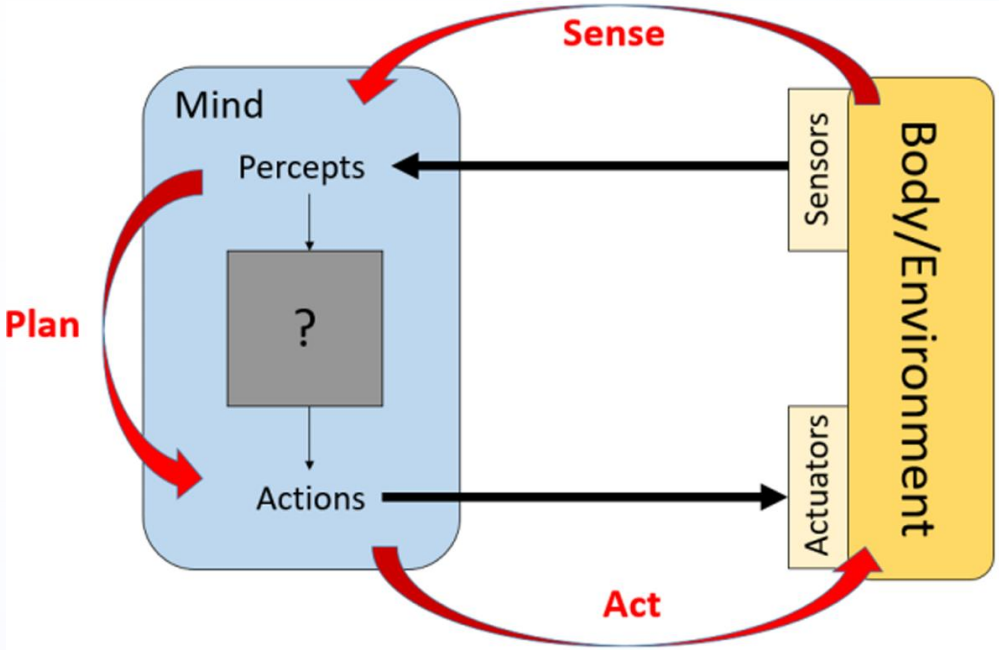
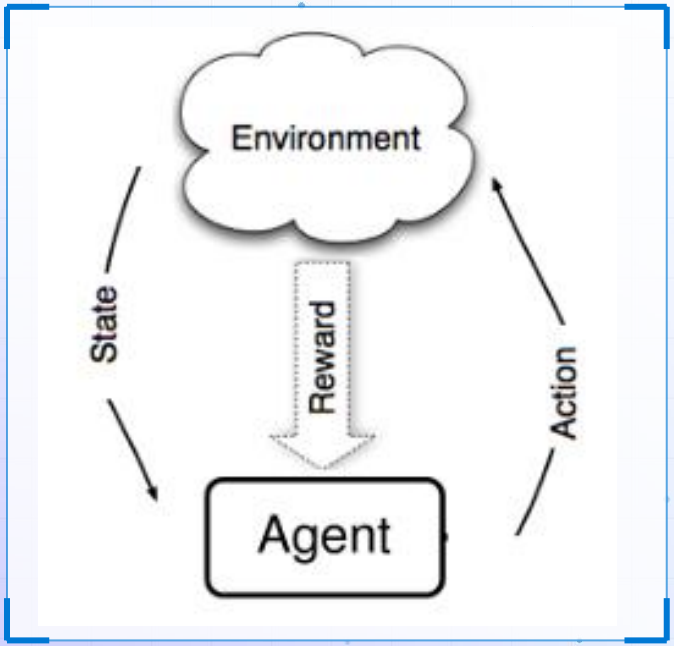


Image extracted from “Robotic Systems” de Kris Hauser, University of Illinois at Urbana-Champaign.

Reinforcement Learning



Agent interacts with the environment through trial and error.

- **Receives an input (perception), with current state “s”.**
- **Then, decides (acts) with an action “a” as output.**
- **Receives a reward (reinforcement) “r”.**

Sequential Decision Problem

state “s” belongs to a Markovian Decision Process if and only if:

$$Pr\{s_{k+1}|s_k\} = Pr\{s_{k+1}|s_1, ..., s_k\}.$$

Reinforcement Learning Problem

S: set of states.

A: set of actions.

$T : S \times A \times S \rightarrow [0, 1]$ transition function (unknown)

R: $S \times A \times S \rightarrow \mathbb{R}$ rewards function

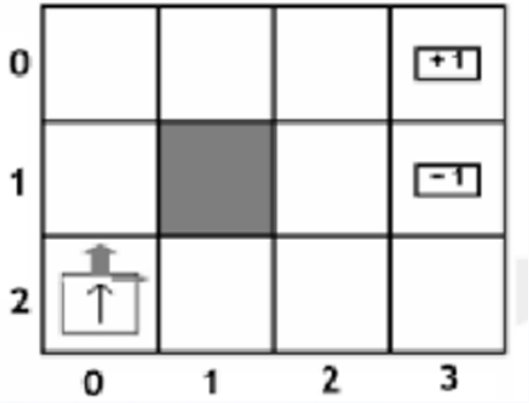
$\pi : S \rightarrow A$ policy

Return $V^{\pi}(s_k) = r_k + \gamma r_{k+1} + \gamma^2 r_{k+2} + ...$

$$= \sum_{i=0}^{\infty} \gamma^i r_{k+i}.$$

Example

Grid-world



States: location on the grid.

Actions: up, down, left, right.

Rewards: +1 on the goal
-1 on prohibited zone
-0.1 otherwise

Basic (and tabular) methods: Value Iteration, Policy Iteration

Value Iteration

0	0.62	0.8	1	+1
1	0.46		0.8	-1
2	0.31	0.46	0.62	0.46
	0	1	2	3

Policy Iteration

0	der.	der.	der.	meta
1	arr.		arr.	prohib
2	arr.	der.	arr.	izq.
	0	1	2	3

Q-learning

Update rule:

$$Q(s_k, a_k) \leftarrow (1-\alpha)Q(s_k, a_k) + \alpha\left(r_{k+1} + \gamma \max_a Q(s_{k+1,a})\right)$$

Then, policy should be given by:

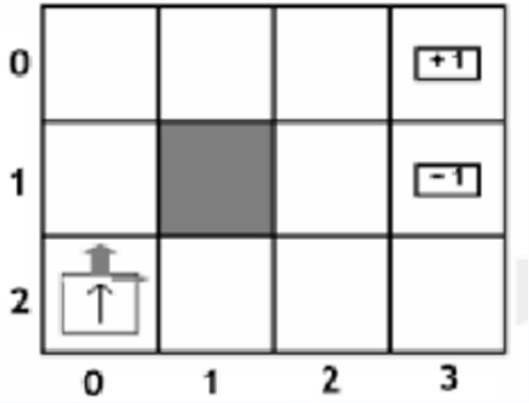
$$\pi^*(s_k) = arg\max Q^*(s_k, a_k)$$



Image extracted from “Introduction to Q-learning with OpenAI Gym”, Gelana Tostaeva, Medium.

Example

Grid-world



States: location on the grid.

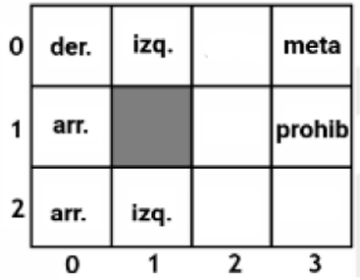
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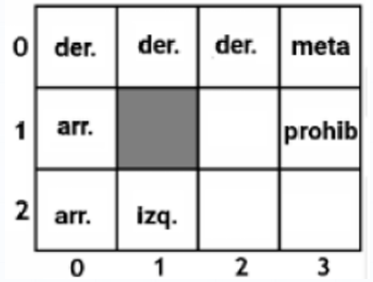
Basic (and tabular) methods: Value Iteration, Policy Iteration

Q-learning

- Visited (state,action) pairs are the only ones updated.

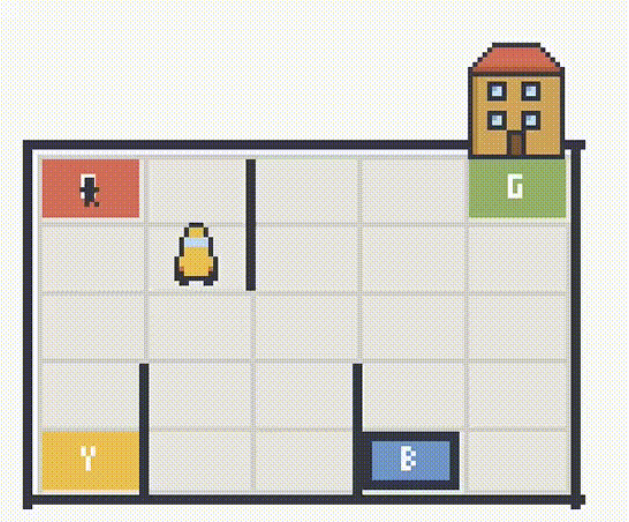


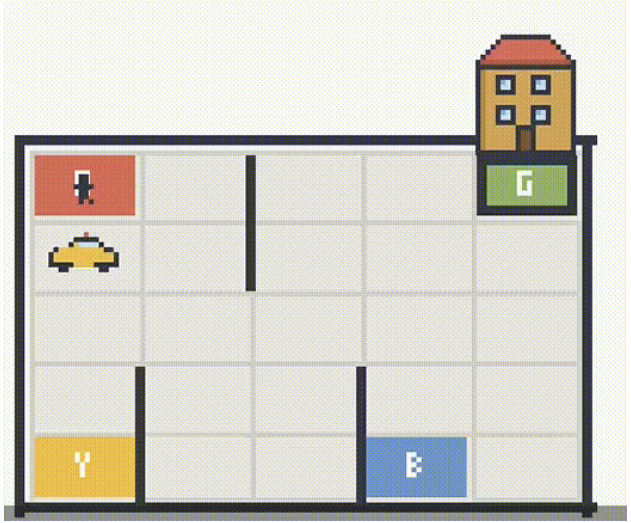
- Including exploration:



Example

Taxi

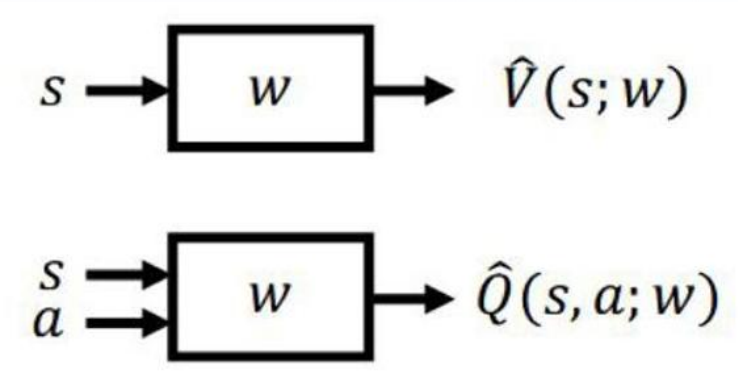




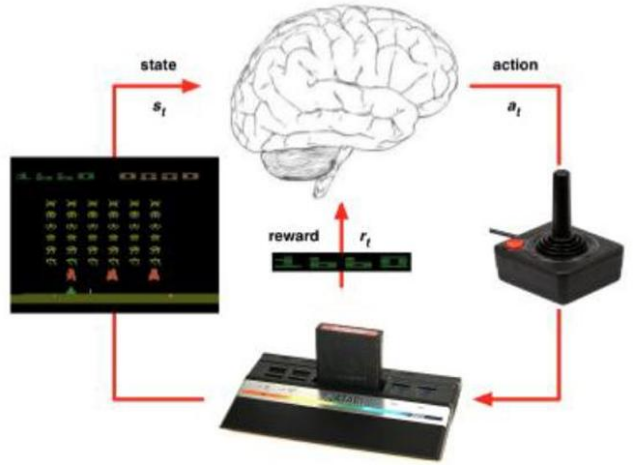
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Deep Reinforcement Learning

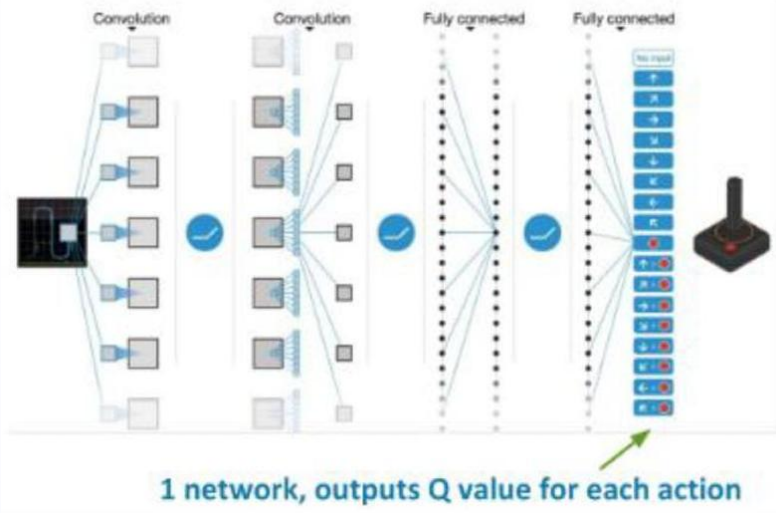
Need of function approximators (such as artificial neural networks)



General scheme (pretty much the same)



Deep Reinforcement Learning



so… discrete action space

Replay buffer

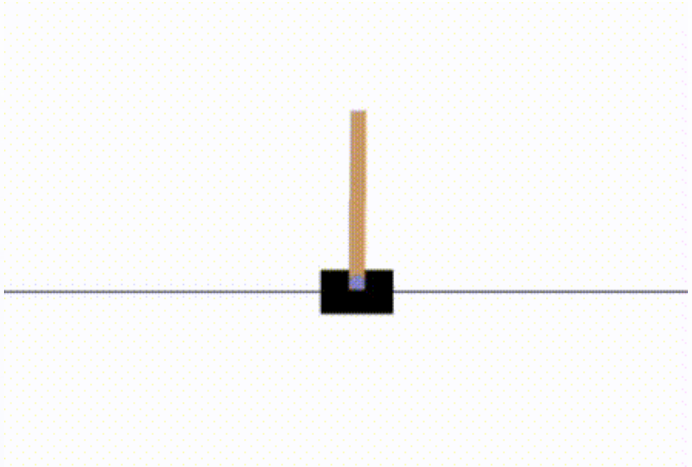
s_1, a_1, r_2, s_2
s_2, a_2, r_3, s_3
s_3, a_3, r_4, s_4
…
$s_t, a_t, r_{t+1}, s_{t+1}$

$\rightarrow \quad s, a, r, s'$

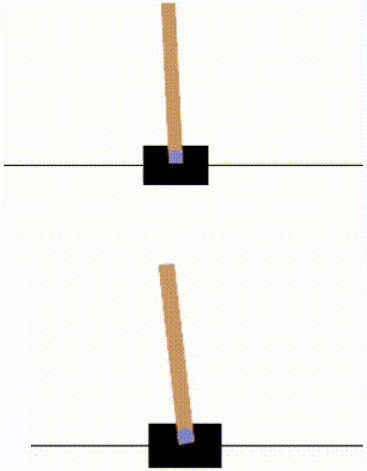
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Deep Reinforcement Learning

Inverted pendulum



Random policies



REINFORCE

Policy parametrization:

$$\Pi = \{\pi_\theta, \theta \in \mathbb{R}^m\}$$

for every policy we would have:

$$J(\theta) = \mathbb{E}\left[\sum_{t\geq 0} \gamma^t r_t | \pi_\theta\right]$$

we always are looking for optimal policy:

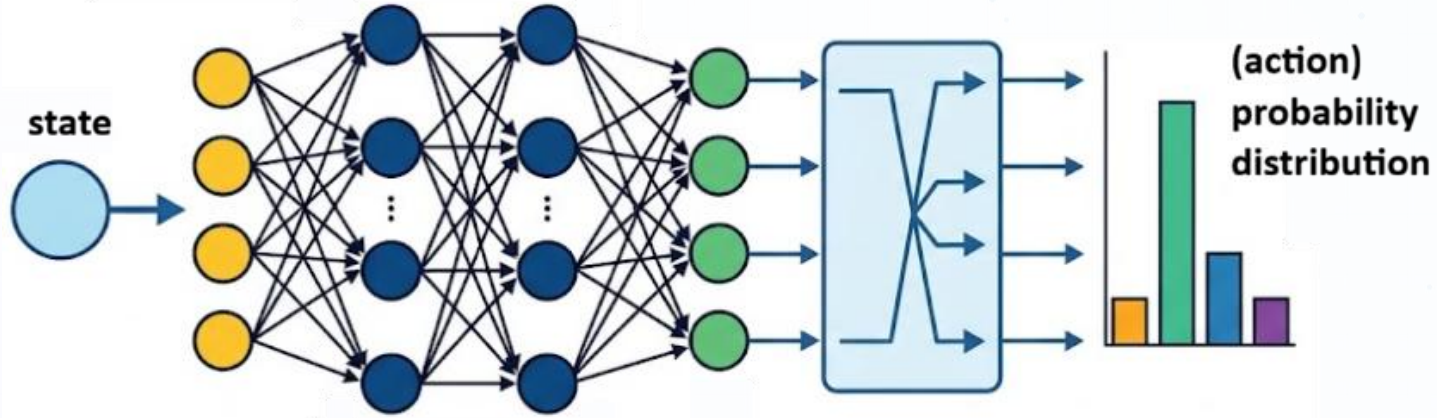
$$\theta^* = \arg\max_\theta J(\theta)$$

but now we have transition probability given by:

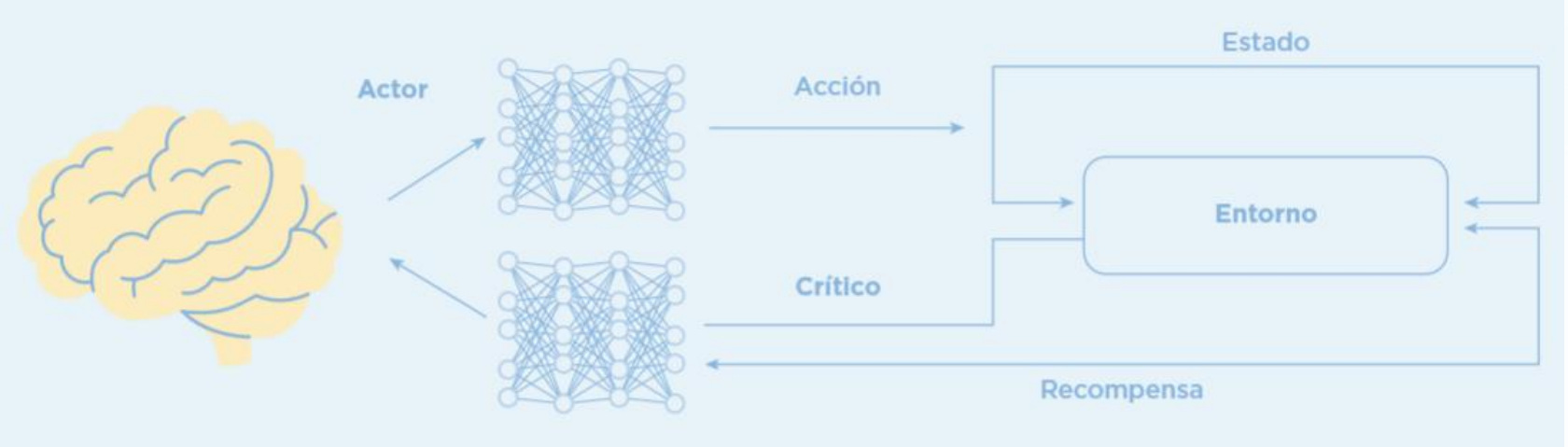
$$p(\tau;\theta) = \prod_{t>0} p(s_{t+1}|s_t, a_t)\pi_\theta(a_t|s_t)$$

REINFORCE

on the hands-on side:

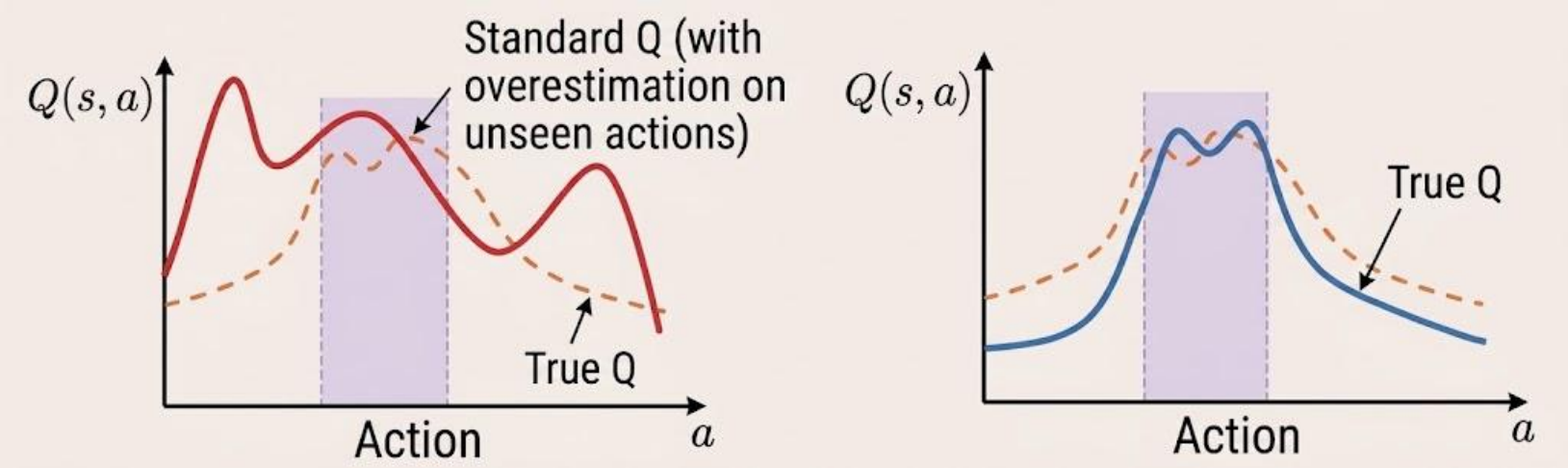


ACTOR-CRITIC



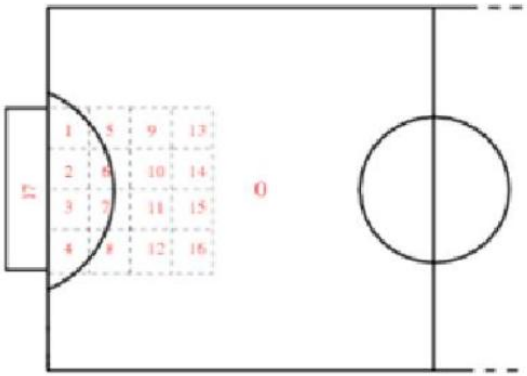
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CONSERVATIVE APPROACHES



APPLICATIONS

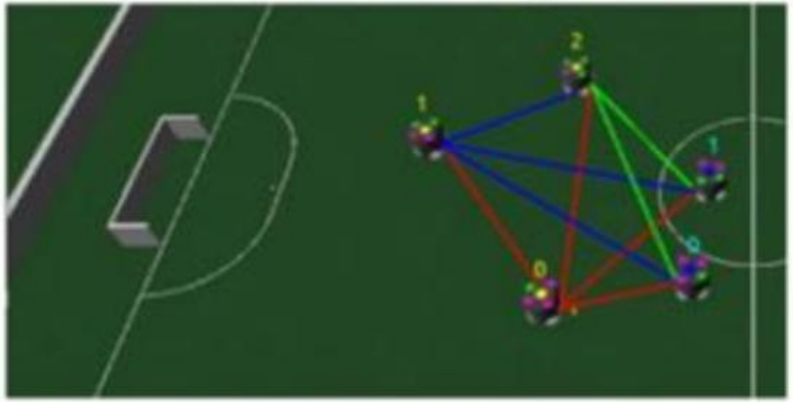




G.A. Ahumada, C.J. Nettle and M.A. Solis, 'Accelerating Q-learning through Kalman Filter Estimations applied in a RoboCup SSL Simulation', **Proceedings** of the 10th IEEE Latin American Robotics Symposium, 2013.

APPLICATIONS

- dist(K_i,pelota), dist(T_j,pelota)
- dist(K_i,K_j)
- dist(K_i,T_j)
- angle(K_i,T_i)

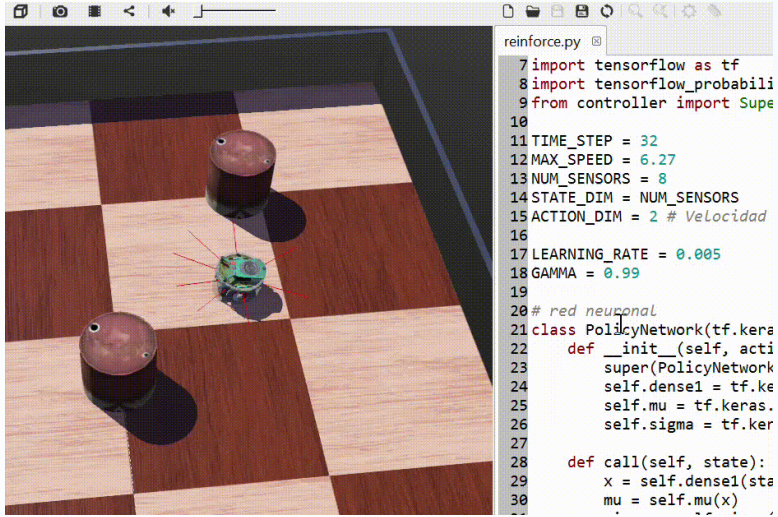


Ollino, F., Solis, M. A., & Allende, H. (2018). Batch reinforcement learning on a RoboCup Small Size League keepaway strategy learning problem. In 4th Congress on Robotics and Neuroscience, CRoNe 2018. CEUR-WS.

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Closing remarks

- ANNs are the most popular function approximators for RL on continuous spaces, but there is room for other techniques (such as fuzzy logic).
- Specific RL algorithm to be applied would depend on the given problem.
- don't miss the workshop tomorrow **Saturday Nov 7th, 9am** at **Library Auditorium**



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// Q&A

Thank you! Questions?

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References

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- Kober, J., Bagnell, J. A., & Peters, J. (2013). Reinforcement learning in robotics: A survey. The International Journal of Robotics Research, 32(11), 1238-1274.
- Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., ... & Hassabis, D. (2015). Human-level control through deep reinforcement learning. nature, 518(7540), 529-533.

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